

DISASTER MANAGEMENT

**CYBER ATTACKS,
CHEMICAL AND
NUCLEAR DISASTERS**

MODULE - 10



THE RECENT SURGE OF CYBER ATTACKS



1

A cyber attack is an attack launched from one or more computers against another computer, multiple computers or networks.

2

Cyber attacks/breaches can be broken down into two broad types:

- Attacks where the goal is to disable the target computer or knock it offline, or
- Attacks where the goal is to get access to the target computer's data and perhaps gain admin. privileges on it.



**CYBERCRIMES WILL COST COMPANIES WORLDWIDE AS THE
GREATEST TRANSFER OF ECONOMIC WEALTH IN HISTORY**



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TYPES OF CYBER ATTACKS

1

MALWARE

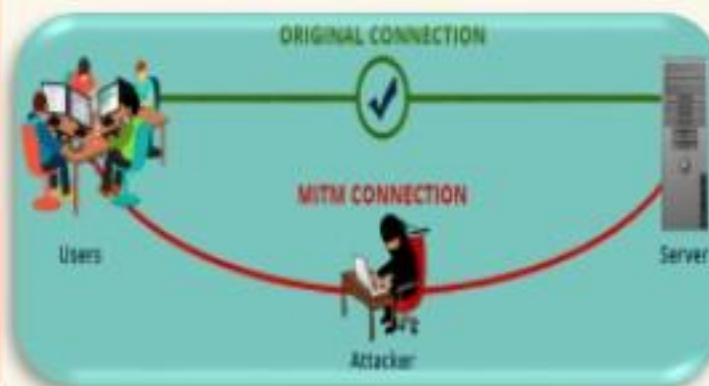
The malware can refer to any kind of software, no matter how it's structured or operated, that is designed to cause damage to a single computer, server, or computer network."



2

MAN IN THE MIDDLE

A man in the middle attack (MITM) is a method by which attackers manage to interpose themselves secretly between the user and a web service they're trying to access.



3

DENIAL OF SERVICE

A denial of service attack is a brute force method to try stop some online service from working properly. It sends so much traffic to a website or so many requests to a database that it overwhelms those systems ability to function



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TYPES OF CYBER ATTACKS

ZERO-DAY EXPLOITS

Zero-days are vulnerabilities in software that have yet to be fixed. Techniques for exploiting such vulnerabilities are often bought and sold on the dark web

CRYPTOJACKING

Cryptojacking is a specialized attack that involves getting someone else's computer to do the work of generating cryptocurrency for you (a process called *mining* in crypto lingo).

PHISHING

Phishing is a technique by which cybercriminals craft emails to fool a target into taking some harmful action.

SQL INJECTION

SQL injection is a means by which an attacker can exploit a vulnerability to take control of a victim's database.



RECENT CYBER ATTACKS

- 1 NOTPETYA, WannaCry were reported ransomware attacks worldwide in 2017
- 2 Ethereum related to bitcoin - Ethereum (ETH) wallet is reportedly injecting malicious javascript code from open browser windows to steal data from its users
- 3 'MedusaLocker' is a ransomware (a type of malware) that makes its way into users' devices by encrypting files.
- 4 Similar Stuxnet virus attack happened on an Iranian nuclear facility in 2018.
- 5 A malware named 'Dtrack RAT' is a new spyware tool that is believed to be actively targeting a large number of Indian institutions, the Nuclear Power Corporation of India Ltd. (NPCIL) has confirmed that a malware had infected its system at the Kudankulam Nuclear Power Plant (KKNPP).
- 6 Whatsapp recently was attacked by a spyware named Pegasus. a technique called steganography is used which can hide malicious code inside a file that appears normal and allow hackers to bypass security software and firewalls.



RECENT EFFORTS TO MITIGATE CYBER ATTACKS

**1**

- Cyberdome is an initiative of Kerala Police Department and also taken up by Assam Police.
- It has an online office called 'Virtual dome' where technical experts and ethical hackers who assist the police in the cyber security.
- c0c0n, also known as the CyOps Con, is an annual event organized as part of the International Information Security Day by the Kerala State Police.



RECENT EFFORTS TO MITIGATE CYBER ATTACKS



2

- Maharashtra became the first state in the country which adopt the AMBIS system.
- The Automated Multi-modal Biometric Identification System (AMBIS) is the new innovation in the way of storing data like fingerprints, photographs, and retinal scans of all criminals.
- This system will help the police in nabbing the culprits in speedy way.

3

Trakea - The Haryana Police Department has adopted a unique barcoding software. The software is used to ensure that 1000s of digital forensic reports are stored.



HOW TO BUILD RESILIENCE AGAINST CYBER ATTACKS?



- 1** **Real-time intelligence** is required for preventing and containing cyber attacks.

- 2** **Securing computing environment:** To achieve resilience against cyberattacks, India needs to secure facilities like Internet of Things (IoT) with current tools, patches, updates in a timely manner.

- 3** **Develop Core skills;** This is the need of the hour is in cyber-security, data integrity, and data security fields and setting up of stringent cyber-security standards to protect the institutional infrastructure of the country.

- 4** **A robust Disaster Recovery and Business Continuity Plan for Critical Infrastructure** so that they can bounce back after disruption.





CENTRAL GOVERNMENT STEPS TAKEN TO DEAL WITH CYBER CRIME AND CYBER SECURITY

- 1** Establishment of **National Critical Information Infrastructure Protection Centre (NCIIPC)** created under Sec 70A of the Information Technology Act, 2000 for protection of critical information infrastructure in the country.
- 2** Cyber And Information Security (C&IS) Division of MHA deals with matters relating to Cyber Security, Cyber Crime, National Information Security Policy & Guidelines (NISPG) and implementation of NISPG, NATGRID etc.
- 3** All organizations providing digital services have been **mandated to report cyber security incidents to CERT-In** expeditiously.
- 4** Issue of guidelines for **Chief Information Security Officers (CISOs)** regarding their key roles and responsibilities for securing applications / infrastructure and compliance.
- 5** **Cyber Swachhta Kendra** (Botnet Cleaning and Malware Analysis Centre) has been launched for providing detection of malicious programmes and free tools to remove such programmes.



CENTRAL GOVERNMENT STEPS TAKEN TO DEAL WITH CYBER CRIME AND CYBER SECURITY



6 Empanelment of security auditing organisations to support and audit implementation of Information Security Best Practices.

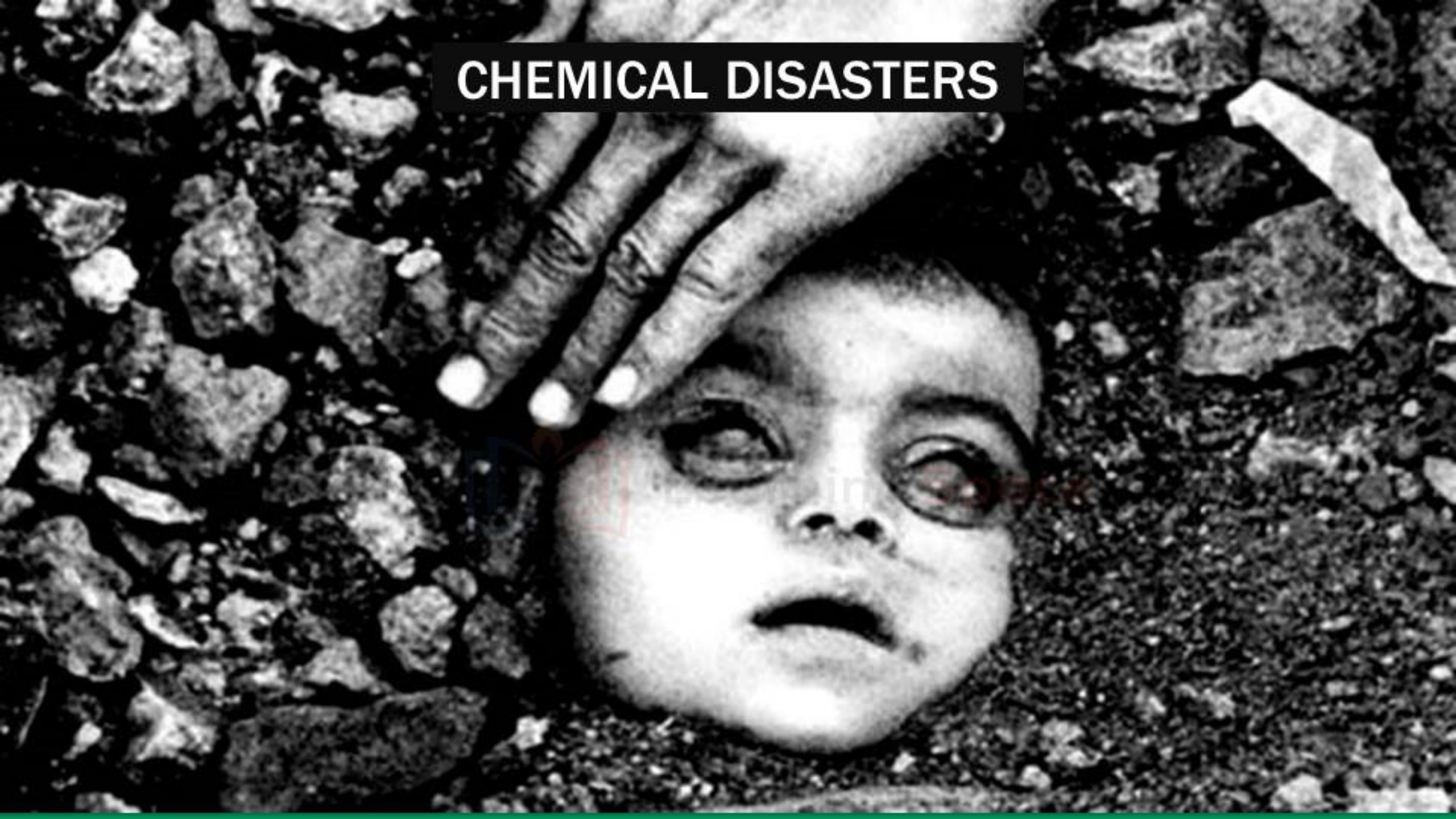
7 Formulation of Crisis Management Plan for countering cyber attacks and cyber terrorism

8 Conducting cyber security mock drills and exercises regularly to enable assessment of cyber security posture and preparedness of organizations in Government and critical sectors.

9 Conducting regular training programmes for network / system administrators and Chief Information Security Officers (CISOs) and regularly testing for vulnerabilities at reasonable intervals of time is very important following a Cyber Crisis Management Plan (CCMP).



CHEMICAL DISASTERS



CHEMICAL DISASTERS

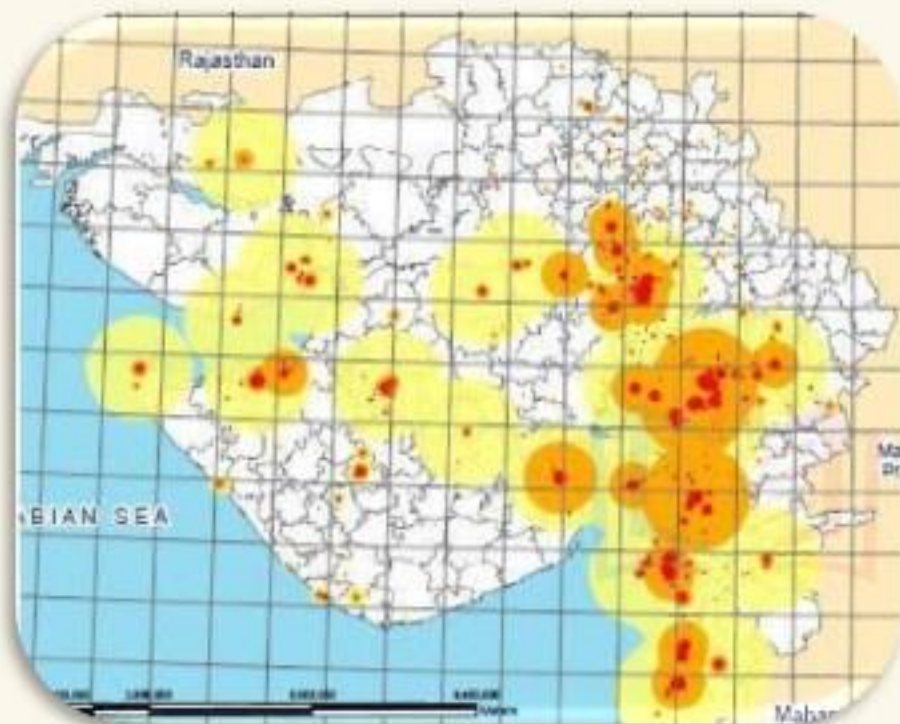


- 1** The growth of industries has led to an increase in the risk of occurrence of incidents associated with hazardous chemicals (HAZCHEM) and hazardous materials (HAZMAT).
- 2** Chemical/ industrial accidents are significant and have long term impact on the community and environment. It leads to injuries, loss of lives, damage to property and environment.
- 3** While the common causes for chemical accidents are deficiencies in safety management , systems or human errors, natural calamities or sabotage may also trigger such accidents.



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CHEMICAL DISASTER RISK IN THE INDIAN STATES



Vulnerable Areas in Gujarat for Toxic Hazards

- There is relatively higher presence of industrial sector along the west coast, largely due to the proximity to raw materials and ports.
- Gujarat alone is estimated to contribute around 53 percent to the total production in the country, followed by Maharashtra, which contributes nine percent.
- The other major producing states include Uttar Pradesh (UP), Tamil Nadu (TN), Madhya Pradesh (MP), and Punjab



CENTRAL RESPONSIBILITY IN UNDERSTANDING RISK OF CHEMICAL DISASTERS

Ministry of Environment, Forest and Climate Change (MOEFCC) is the nodal ministry for All types of Chemical Disasters.

SHORT TERM

Online information system on HAZCHEM conforming to international standards
Chemical Accident Information Reporting System
Information on dealing with HAZCHEM.

MEDIUM TERM

Research on effective management of HAZCHEM, National Hazardous Waste Information System (NHWIS) gives the status of Hazardous Waste Management in the country.

LONG TERM

Promote research and studies – both in-house and extra-mural by providing research grants to researchers and institution, Promote R&D for indigenous manufacture of quality personal protection equipment, most of which are currently imported



RELEVANT PROVISIONS ON CHEMICAL DISASTER MANAGEMENT, PREVAILING IN COUNTRY

Disaster Management Act 2005

Petroleum Act 1934

Public Liability Insurance Act 1991

Factories Act 1948

Motor Vehicles Act 1988

Insecticides Act 1968

Explosives Act 1884

Environment Protection Act 1986



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HUMAN INDUCED DISASTERS

NUCLEAR AND RADIATION EMERGENCIES (NRE)



- 1 A nuclear disaster is construed as potentially a low probability event, however very high in damage impact.

- 2 It could be **caused** by detonation of nuclear warhead (NWH) or explosion of an Improvised Nuclear Device (IND) or failure of a Nuclear Reactor.

- 3 It is associated with release of large amounts of devastating energy due to Blast, Thermal and Radioactive material.

- 4 The Secondary effects occurring later might result in fall out of radioactive dust.



The International Atomic Energy Agency (IAEA) classifies the NRE emergency scenarios under two broad categories

NUCLEAR EMERGENCIES

- A nuclear emergency is a hazard **due to the release of energy along with radiation** from a nuclear chain reaction (or from the decay of the products of a chain reaction).
- These covers accidents in nuclear reactors, 'criticality' situations in fuel cycle facilities, nuclear explosions, etc.

RADIOLOGICAL EMERGENCIES

- All other emergency situations which have the potential hazard of **radiation exposure due to decay of radioisotopes**, are classified as radiological emergencies.
- They may range from an isolated occupational or medical over-exposure of a person, to a major catastrophe with global dimensions.



CLASSIFICATIONS OF NRE EMERGENCIES

- 1** A 'criticality' accident in a nuclear fuel cycle facility where an uncontrolled nuclear chain reaction takes place inadvertently leading to bursts of neutrons and gamma radiation.
- 2** An accident involving radioactive material package during its transportation.
- 3** The malevolent use of Radiological Dispersive Devices (RDD) or Improvised Nuclear Devices (IND) by terrorists.
- 4** An accident taking place in any nuclear facility of the nuclear fuel cycle including the nuclear reactor, or in a facility using radioactive sources, leading to a large-scale release of radioactivity in the environment.
- 5** A large-scale nuclear disaster resulting from a nuclear weapon attack, which would lead to mass casualties and destruction of large areas and properties.



RISK REDUCTION FOR NUCLEAR EMERGENCIES

**Department of
Atomic Energy,
Ministry of Home
Affairs, Ministry Of
Defence**

Short Term - Establish set ups for monitoring, warning including Indian Environmental Radiation Monitoring Network (IERMON)

Medium Term - Strengthening radiation Monitoring and Detection Systems in the Public Domain . Install radiation detectors at all identified locations at border posts, and ports.

India's first indigenous anti-nuclear medical kit developed by the Institute of Nuclear Medicine and Allied Sciences (INMAS) that are previously procured from USA.



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LONG TERM RISK MANAGEMENT OF NUCLEAR WASTE

- 1** Current approaches of management:
 - Segregation and storage for short-lived waste,
 - Near-surface disposal for low and some intermediate-level waste,
 - Burial in a deep geological repository or transmutation for the high-level waste.
- 2** According to United Nations Institute for Disarmament Research UNIDR- Nuclear risk reduction from the nuclear weapons has been the talks between governments and presented a report on Nuclear Risk Reduction: A Framework for Analysis.
- 3** Managing 'Radioactive Waste' produced at all stages of the Nuclear Fuel Cycle
 - The fuel cycle involves the mining and milling of uranium ore,
 - The processing and fabrication into nuclear fuel,
 - Its use in the reactor,
 - Its reprocessing (if conducted),
 - The treatment of the used fuel taken from the reactor, and finally,
 - The disposal of the waste.





RELATED QUESTIONS THAT CAN BE ANSWERED AFTER READING THE MODULE



- 1** What is a cyber breach and what is the vulnerability status of India towards it? With respect to the latest reports of attacks describe how the events can be more efficiently handled?
- 2** India sees one of the highest number of cyber attacks and the country ranks second in terms of targeted attacks. How the recent threats can be a prudent way of a robust cyber surveillance in the country.
- 3** What are the causes of chemical disasters? Why is the west coast of India more vulnerable to the chemical disasters than other parts of India?
- 4** Distinguish between nuclear emergencies and radiological emergencies. How can the risk of nuclear waste be identified and reduced in long term?



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